

Notes on the End-effector velocity and forces, MLS Chapter 3, Sections 4.1 & 4.2

NJ Theron

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On the twist coordinates that form the columns of the spatial manipulator Jacobian, eq. (3.52) p. 115

The implication of eq. (3.52) p. 115 is that each of the 4×4 matrices

$$\frac{\partial g_{st}}{\partial \theta_i} g_{st}^{-1}$$

is of a form such that

- its 4th row contains 4 zero elements and
- its 3×3 upper left sub-matrix is skew-symmetric,

so that this matrix can be unpacked into a 6×1 column vector

$$\left(\frac{\partial g_{st}}{\partial \theta_i} g_{st}^{-1} \right)^v.$$

This was never shown, perhaps because it is actually quite easy to show this.

On pp. 51 & 52, in Lemma 2.12, MLS prove for a rotation matrix $R(t) \in SO(3)$ the matrix $\dot{R}(t)R(t)^T$ is skew-symmetric. The proof of this lemma is based on the fact that $R(t)R(t)^T = I$. By exactly the same argument as used in Lemma 2.12, one can show that for the rotation matrix $R(\theta) \in SO(3)$, the matrix

$$\frac{\partial R(\theta)}{\partial \theta_i} R(\theta)^T$$

is skew-symmetric, because $R(\theta)R(\theta)^T = I$.

On p. 54 MLS prove for a transformation matrix $g_{ab}(t) \in SE(3)$ the matrix $\dot{g}_{ab}g_{ab}^{-1}$ is a 4×4 matrix with its 4th row filled with zero elements and its 3×3 upper left sub-matrix is skew-symmetric matrix, $\dot{R}_{ab}R_{ab}^T$. By exactly the same argument as used on p. 54, one can show that for the transformation matrix $g_{st}(\theta) \in SE(3)$, the matrix

$$\frac{\partial g_{st}}{\partial \theta_i} g_{st}^{-1}$$

is a 4×4 matrix with its 4th row filled with zero elements and its 3×3 upper left sub-matrix is skew-symmetric matrix,

$$\frac{\partial R_{st}(\theta)}{\partial \theta_i} R_{st}(\theta)^T .$$

This therefore implies that this 4×4 matrix is indeed a twist $\hat{\xi} \in \mathbb{R}^{4 \times 4}$, which can be expressed in twist coordinates

$$\left(\frac{\partial g_{st}}{\partial \theta_i} g_{st}^{-1} \right)^\vee \in \mathbb{R}^6 .$$

On deriving eq. (3.53) p. 116

The product of exponentials formula is

$$g_{st}(\theta) = e^{\hat{\xi}_1 \theta_1} e^{\hat{\xi}_2 \theta_2} \dots e^{\hat{\xi}_{i-1} \theta_{i-1}} e^{\hat{\xi}_i \theta_i} \dots e^{\hat{\xi}_n \theta_n} g_{st}(0) \quad (3.3)$$

(also shown, unnumbered, a few lines above eq. (3.53)). Premultiplying both sides of eq. (3.3), first by $e^{-\hat{\xi}_1 \theta_1}$, then the resulting equation by $e^{-\hat{\xi}_2 \theta_2}$, then by $e^{-\hat{\xi}_3 \theta_3}$, and so forth, renders:

$$e^{-\hat{\xi}_{i-1} \theta_{i-1}} \dots e^{-\hat{\xi}_3 \theta_3} e^{-\hat{\xi}_2 \theta_2} e^{-\hat{\xi}_1 \theta_1} g_{st}(\theta) = e^{\hat{\xi}_i \theta_i} \dots e^{\hat{\xi}_n \theta_n} g_{st}(0) . \quad (a)$$

When MLS calculates $(\partial g_{st} / \partial \theta_i) g_{st}^{-1}$ just above eq. (3.53) and completes the second of the three lines, they then substitute the matrix factors

$$e^{\hat{\xi}_i \theta_i} \dots e^{\hat{\xi}_n \theta_n} g_{st}(0)$$

to the right of the matrix factor $\left(\hat{\xi}_i \right)$, which is also the right hand side of the equation (a) above, with the left hand side of equation (a). This results in string of matrix factors, of which the last two are $g_{st} g_{st}^{-1}$, equal to a 4×4 identity matrix. This therefore renders the third line of the three-line equation, from which then follow eq. (3.53), by application of lemma 2.13 p. 56.

On the 2nd last equation on p. 117

Note that implicit in the second of the three expressions for the velocity of the tool frame

$$R_{st} \hat{V}_{st}^b \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

is that only the 1st 3 rows of the 4×4 matrix $\hat{V}_{st}^b$ are used in the calculation, since R_{st} is 3×3 .

On new notation used in Example 3.9 p. 119

In Example 3.9 p. 119 MLS introduce a new notation (not been used earlier in the book). $\hat{x}$ and $\hat{z}$ should be interpreted as the unit vectors in the spatial frame positive x and z axis directions. In this example, therefore,

$$\hat{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \text{and} \quad \hat{z} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} .$$

On Example 3.10 p. 123

When MLS, in doing example 3.10, states on p. 123 that the null space of the matrix $(J_{st}^s)^T$ is spanned by the column vectors F_{N_1} and F_{N_2} , they are saying that

$$(J_{st}^s)^T F_{N_1} = 0 \quad \text{and} \quad (J_{st}^s)^T F_{N_2} = 0 ,$$

that is, 4×1 zero matrices. Since both F_{N_1} and F_{N_2} are spatial wrenches, this means that, by eq. (3.60) p. 121, the joint torques τ are zero in both cases where the end effector is loaded by wrenches F_{N_1} and F_{N_2} , which then leads to their conclusion.